Lab Guide:

GCP Cloud Account Onboarding to SCM AIRS

Overview

This lab guides you through the process of onboarding a Google Cloud Platform (GCP) account to the **Strata Cloud Manager (SCM) AI Runtime Security (AIRS)** dashboard. By completing these steps, you will enable SCM to discover unprotected AI and non-AI traffic and models within your GCP infrastructure.

Prerequisites

Before starting this onboarding process, ensure you have:

- Completed the **Phase 1** infrastructure deployment via Terraform.
- The name of the **GCS Storage Bucket** created during Phase 1 (Format: `dc-master-<Userid>`).
- The **GCP Project ID** where the assets are located.
- Terraform installed on your local machine.

Technical Readiness Checklist

To ensure the Terraform execution in Step 4 is successful, verify the following configuration in your GCP project:

1. Enable Required APIs

Ensure the following permissions are enabled for the service account used for deploying terraform:

- `cloudasset.assets.listResource`
- `cloudasset.assets.listAccessPolicy`
- `cloudasset.feeds.get`
- `cloudasset.feeds.list`
- `compute.machineTypes.list`
- `compute.networks.list`
- `compute.subnetworks.list`
- `container.clusters.list`
- `pubsub.subscriptions.consume`
- `pubsub.topics.attachSubscription`
- `storage.buckets.list`
- `aiplatform.models.list`

2. Create GCP Service Identity

Run the following command in your terminal to create the necessary service identity for Cloud Asset Inventory. Replace <GCP_PROJECT_ID> with your project ID:

```
gcloud beta services identity create --service=cloudasset.googleapis.com --
project=<GCP_PROJECT_ID>
```

Note : Click Here to check full pre-requisites to successful GCP cloud account onboarding. VPC flow and audit logs enablement & storage to GCS bucket is taken care by the Phase-1 terraform run, so in case you do hit any issues with discovery terraform script, then recheck the IAM permissions for the terraform user.

3. Verify Log Routing

Note : Deployed as part of terraform run for cloud Infra build in phase 1.

- **VPC Flow Logs:** Ensure flow logs are enabled on your workload subnets with a 5-second aggregation interval and 100% sample rate.
- **Log Sink:** Confirm a Log Router sink is configured to send `vpc_flows` and `cloudaudit.googleapis.com` logs to your GCS bucket (`dc-master-<Userid>`).

Step 1: Access the Cloud Account Manager

1. Log in to your **SCM Tenant**.
2. Navigate to **AI Security** > **AI Runtime Firewall**.

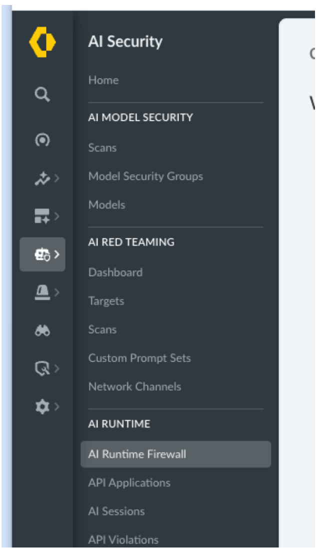


Figure 1: Navigating to the AI Runtime Firewall section in SCM.

3. Click the **Cloud Icon** (Cloud Account Manager) located in the top-right corner of the dashboard.

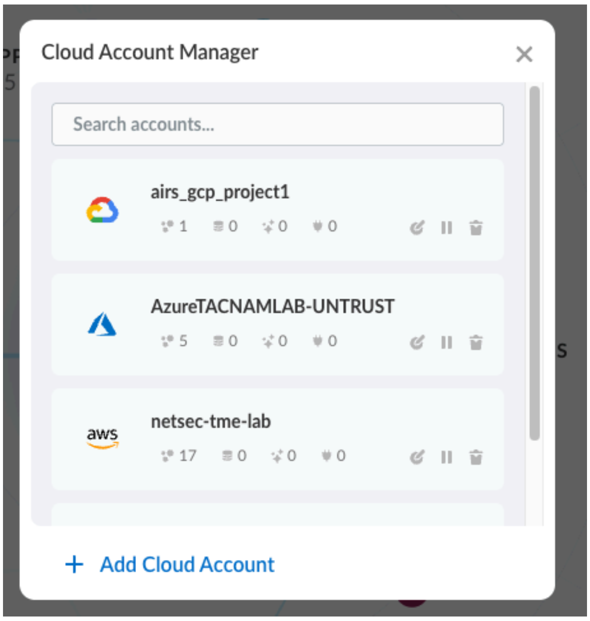


Figure 2: Locating the Cloud Account Manager icon.

Step 2: Configure Cloud Account Details

1. Click **Add Cloud Account** to launch the onboarding workflow.
2. Select GCP as the cloud provider.
3. Enter a unique **Account Name** and your **GCP Project ID**.

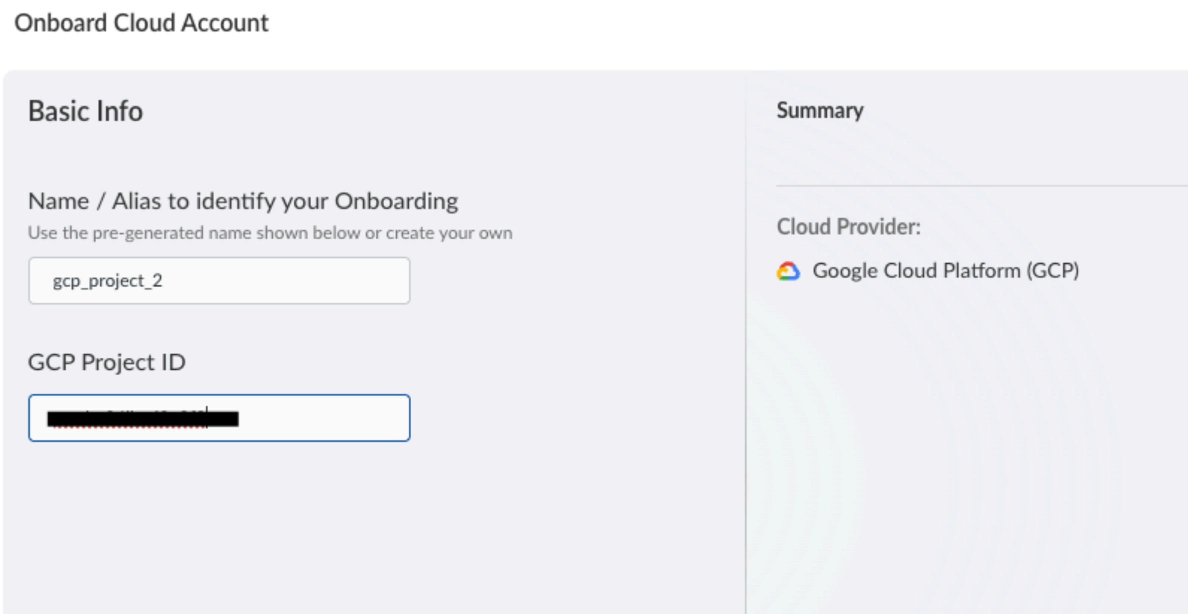


Figure 3: Entering the GCP Project ID and Account Name.

3. Click **Next**.

4. Review the requested permissions and click **Next** to proceed to the Application definition.

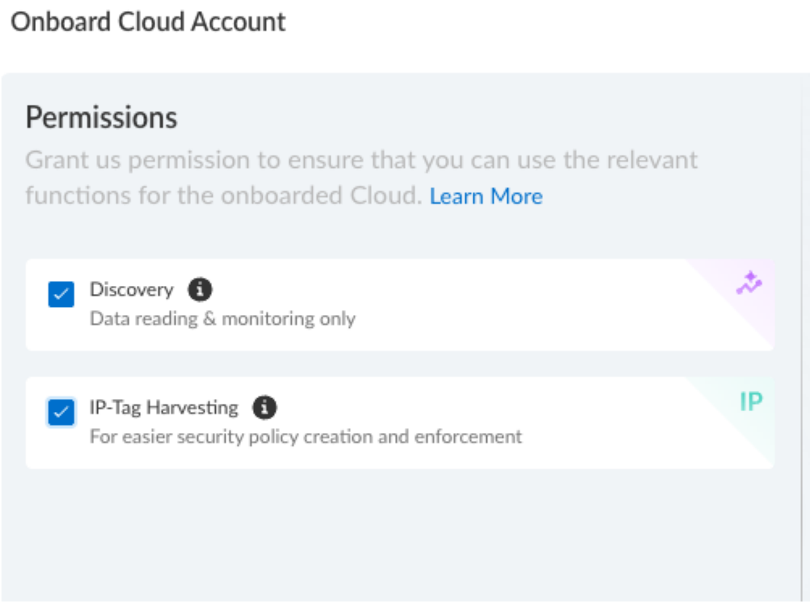


Figure 4: Reviewing the required onboarding permissions.

Step 3: Configure Storage and Service Accounts

1. Leave the **Application definition** as the default setting.
2. In the **Storage bucket Name** field, enter the name of the GCS bucket created during your Phase 1 Terraform deployment (e.g., `dc-master-<Userid>`).
    - *Note: You can verify this name in the outputs of your previous Terraform run.*

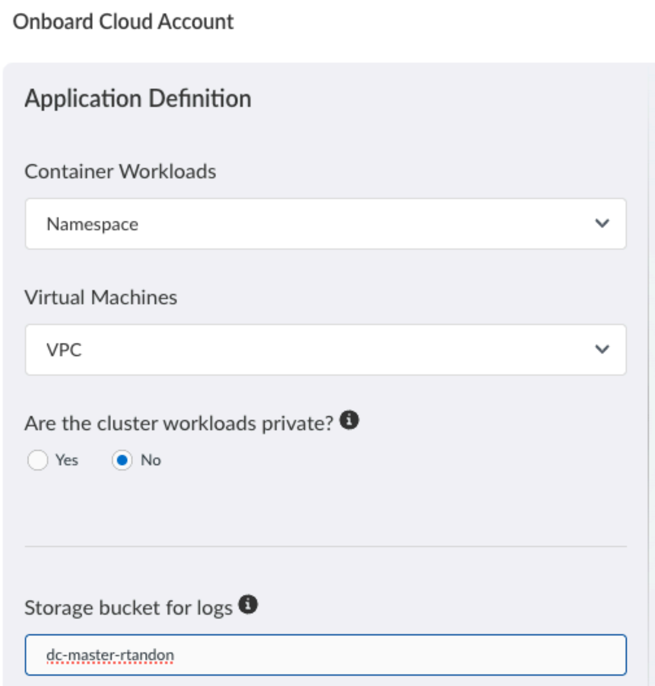


Figure 5: Configuring the GCS storage bucket name.

3. Provide a **Service Account Alias**. This name will be utilized in the Terraform script generated by SCM.

Step 4: Execute Onboarding Terraform Script

1. Click **Download Terraform** to save the configuration ZIP file to your machine.

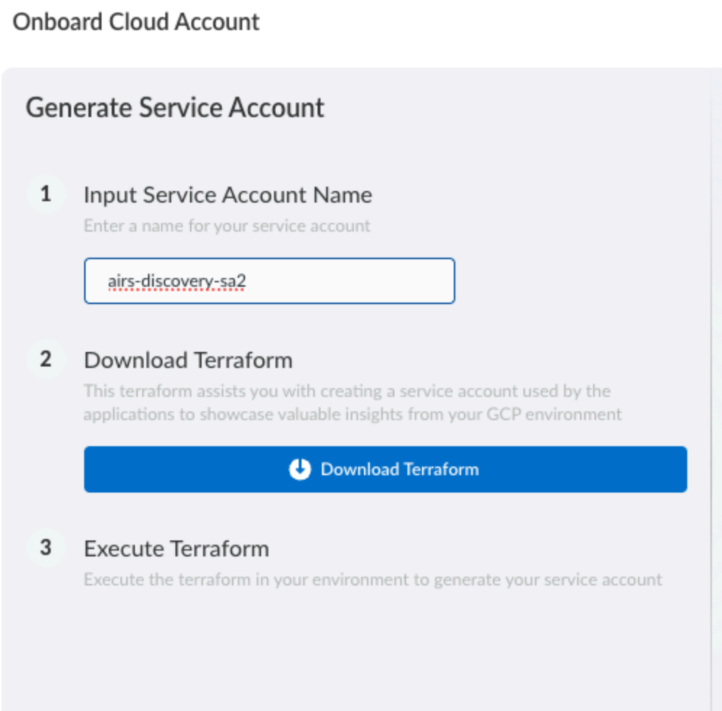


Figure 6: Downloading the generated Terraform package.

2. Unzip the file and open your terminal (macOS) or Command Prompt (Windows) in that directory.
3. Run the following commands to provision the necessary IAM roles and services:
4. `terraform init`
5. `terraform plan`
6. `terraform apply`

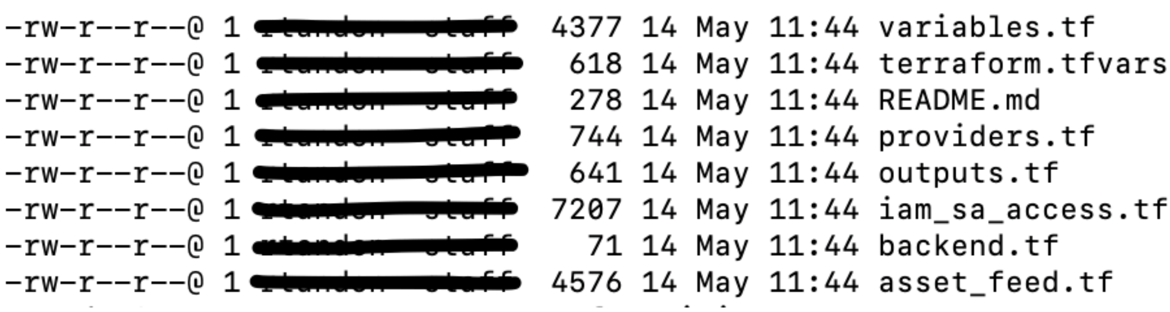


Figure 7: Running Terraform commands in the terminal.

4. Confirm the ‘terraform apply’ when prompted. This script performs the following actions:
   - Creates service accounts for discovery, orchestration, and route stitching.
   - Establishes trust by allowing Palo Alto Networks service accounts to impersonate local service accounts.
   - Sets up the **Cloud Asset Inventory** feed and **Pub/Sub** for real-time monitoring.
   - Enables required GCP APIs (Cloud Asset and Billing APIs).

Step 5: Verify Discovery

1. Once the Terraform execution completes successfully, return to the SCM dashboard and click **Done**.

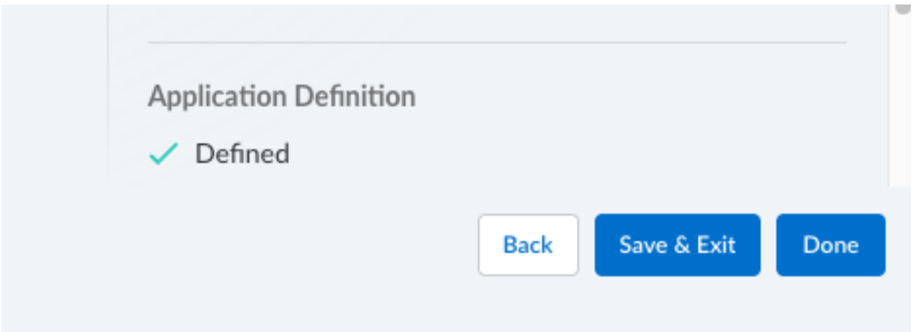
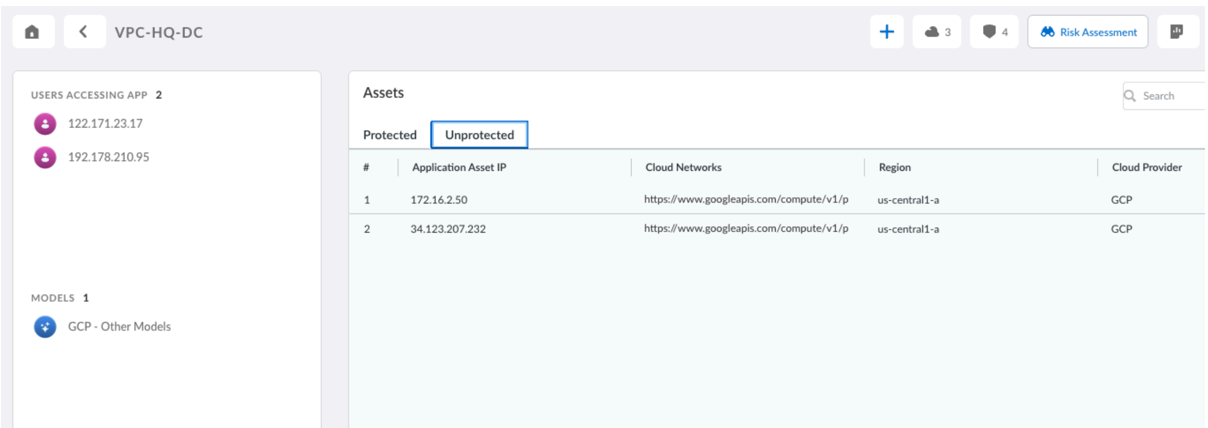


Figure 8: Clicking Done to finalize the workflow in SCM.

2. **Generate Traffic:** Interact with the Streamlit AI application to initiate traffic.
3. **Wait for Discovery:** Allow 10–15 minutes for SCM to complete the initial discovery scan.
4. Navigate to the **Unprotected** section of the SCM AIRS Dashboard. You should now see data populating, showing traffic flows from users to applications and models.



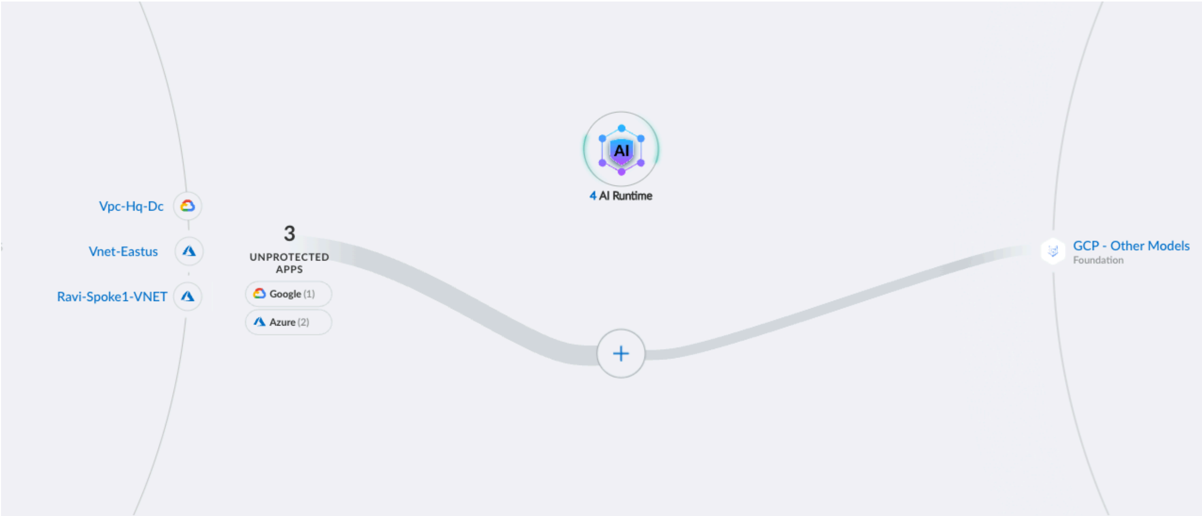


Figure 9: Viewing discovered traffic and models in the AIRS dashboard.

Step 6: Deploy Phase 2 (AIRS Protected)

Click open '1.AIRS_Masterclass_Lab_Guide' and navigate to Phase2 section.
Follow the steps to deploy deploy AIRS firewall to intercept the AI traffic between AI Server and GCP Vertex AI models.

Result: Your SCM is now configured to discover and monitor your GCP resources in real-time through secure service account impersonation. You have now completed Phase1 section of the lab in which you have created a unprotected Cloud infra which hosts an AI application which uses Vertex AI (LLM Provider service). You are now ready to move to Phase 2 in which you will deploy AIRS firewall to intercept the AI traffic between AI Server and GCP Vertex AI models.